

Coin.h Copy-paste code

```
//Coin.h
//DO NOT MODIFY THIS SECTION
#ifndef CLASSNAME_H
#define CLASSNAME_H
#include <iostream>
class Coin{
private:
    std::string sideup;
public:
    Coin() //constructor
    { toss(); }
    std::string getSideUp() //accessor
    { return sideup; }
    void toss();
};
#endif // CLASSNAME_H
```

CoinFlip.cpp copy-paste code

```
#include <iostream>
#include "Coin.h"
using namespace std;

int main() {
    srand(time(0));
    int numHeads = 0, numTails = 0;
    Coin c1;

    for (int i = 0; i < 10; i++) {
        c1.toss();
        cout << c1.getSideUp() << endl;
        if (c1.getSideUp() == "heads") {
            numHeads++;
        }
        else {
            numTails++;
        }
    }
    cout << "Number of heads: " << numHeads << endl;
    cout << "Number of tails: " << numTails << endl;
    return 0;
}

void Coin::toss() {
    int side = rand() % 2;

    if (side == 0) {
        sideup = "heads";
    } else {
        sideup = "tails";
    }
}
```

CoinGame.cpp copy-paste code

```
#include <iostream>
#include "Coin.h"
using namespace std;

int main() {
    srand(time(0));
    Coin quarter, dime, nickel;
    double balance = 0.00;

    while (balance <= 1.00) {
        quarter.toss();
        dime.toss();
        nickel.toss();
        if (quarter.getSideUp() == "heads") {
            balance += 0.25;
        }

        if (dime.getSideUp() == "heads") {
            balance += 0.10;
        }

        if (nickel.getSideUp() == "heads") {
            balance += 0.05;
        }
        cout << "Balance: " << balance << endl;
    }

    if (balance > 1.0) {
        cout << "\nYou lost" << endl;
    } else {
        cout << "\nYou won!" << endl;
    }
    return 0;
}

void Coin::toss() {
    int side = rand() % 2;

    if (side == 0) {
        sideup = "heads";
    }
    if (side == 1) {
        sideup = "tails";
    }
}
```

Output for CoinFlip.cpp

```
tails
tails
heads
tails
heads
heads
Number of heads: 5
Number of tails: 5|

Process finished with exit code 0
```

Output 2 for CoinFlip.cpp

```
tails
tails
heads
tails
tails
heads
Number of heads: 3
Number of tails: 7

Process finished with exit code 0
```

Output for CoinGame.cpp

```
Balance: 0.1  
Balance: 0.25  
Balance: 0.65  
Balance: 0.9  
Balance: 0.9  
Balance: 1.15  
  
You lost  
  
Process finished with exit code 0
```

Output 2 for CoinGame.cpp

```
Balance: 0.25  
Balance: 0.6  
Balance: 0.85  
Balance: 0.95  
Balance: 1.3  
  
You lost  
  
Process finished with exit code 0
```